SEPARATE PROTOTYPE: passive Rd; no CC decoder or new1.5A source restriction. 500mA engineering basis; not USB qualification.

USB main -> mux -> buck. Charger branch ILIM=IN50..100mA; nominal25mA. Boost fed ONLY by15F cap.

Independent resistor-limited raw AON. Every power/control path captured. INPUT_FAULT_N stays local; MCU translator NOT implemented.

Effective-C, input transients, bootstrap/boost PFM/current, cap ESR and thermal are BENCH GATES; not manufacturing release.

Sheet: /
File: separated-candidate.kicad_sch

Title: NOVA separated power prototype – main

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FAULT SR latch holds all converter/load enables off after switch fault; raw–AON remains powered until external cycle.

CAP SR latch arms only with qualified USB and cap–start. Loaded–cap UVLO or proven backup loss disarms; cap rebound cannot rearm.

No status blanking. USB status senses RAW + LIMITED_5V, never the cap–fed common rail alone.

Schmitt buffers condition all open–drain monitor inputs. Source/power–off/propagation timing remains physical validation.

Sheet: /Control / real components/
File: control.kicad_sch

Title: NOVA separated power prototype – control

Size: A1
Date:
KiCad E.D.A. 10.0.7
Rev: SEP–PROTOTYPE–1
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